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***B.Tech. Degree I Semester Regular/Supplementary and
II Semester Supplementary Examination March 2022***

**CE/EE 19-200-0103A ENGINEERING GRAPHICS
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A

(1 × 12 = 12)

- I. A point on a rectangular hyperbola is 30 mm from one of the asymptotes and 20 mm from the second asymptote. Draw the curve and also draw a tangent and normal to the curve. (12)

OR

- II. On a map, a line measuring 1.5 cm represents the actual distance of 1.2 m. Draw a diagonal scale showing the smallest division of 5 cm and long enough to measure up to 16 m. Show a distance of 11.75 m. (12)

PART B

(4 × 15 = 60)

- III. Point A of straight-line AB is 25 mm above HP and 40mm in front of VP. Point B is 30 mm below HP and 15 mm behind VP. Find the true length and inclinations of the line if the end projectors coincide. Also show the traces. (15)

OR

- IV. A circular lamina of 50 mm diameter appears to be an ellipse of major axis 50 mm and minor axis 30 mm in the front view. Draw its projections. (15)

- V. A square pyramid, base 40 mm side and axis 75 mm long is placed on the ground on one of its base edges, inclined 60 degrees to VP, while the axis is inclined 45 degrees with the H.P. Draw its projections. (15)

OR

- VI. A cylinder, 65 mm diameter and 90 mm long is resting on HP, has its axis parallel to the H.P. It is cut by a vertical section plane inclined 30 degrees to HP and perpendicular to VP. Draw its sectional top view and true shape of the section. (15)

- VII. A cone made up of Aluminium sheet with base circle diameter 65 mm and axis length 75 mm is kept with its base on the ground. A circular hole of 30 mm diameter is cut through the cone such that its axis remains perpendicular to V.P.; 10 mm to the right of the axis of cone and 25 mm above the base of cone. Develop the surface of the cone. (15)

OR

- VIII. A square prism of base side 60 mm rests on one of its ends on the HP with the base sides equally inclined to the VP. It is penetrated fully by another square prism of base side 45 mm with the base side equally inclined to the HP. The axes intersect at right angles. The axis of the penetrating prism is parallel to both the HP and the VP. Draw the projections of the prisms and show the lines of intersection. (15)

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- IX. A hexagonal prism of base edge, 20 mm and height 60 mm rests centrally on the top of a cube of side 40mm. Draw the isometric view of the combinations. (15)

OR

- X. A pentagonal pyramid axis length 50 mm and base edge 25 mm is resting on GP on its base. One of the base edges which is nearer to PP is parallel to it and 15mm behind it. The SP is 65 mm above GP and 40 mm in front of PP. Draw the perspective view of the pyramid if the axis is lying on the central plane and the pyramid is completely behind the PP. (15)
